

Affan Rafiuddin

SR.SOLAR ENGINEER

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PROFILE

Reliable Engineer with 6 years of experience in the Renewable Energy and Mechanical Engineering sectors. Skilled in Technical, Management, Engineering, and Strategic Planning. Strong Operations Professional with hands-on experience in the successful execution and coordination of large-scale solar projects exceeding 10 MW, with expertise in project operations, site coordination, execution, and cross-functional team management. ; Saudi Council of Engineers Member ID: 1095544

PROFFESIONAL EXPERIENCE

2025 – Present

Sr. Solar Engineer — OMAS — Jeddah, Saudi Arabia

- Coordinate engineering, procurement, construction, and site teams to ensure efficient execution of solar PV projects.
- Monitor project progress, schedules, quality, technical requirements, and key deliverables to support timely project completion.
- Coordinate technical requirements, site activities, procurement needs, inspections, and project documentation across multidisciplinary teams.
- Prepare and review progress reports, schedules, technical documentation, inspection records, and project correspondence.
- Ensure project execution complies with engineering standards, quality requirements, HSE procedures, and client specifications.
- Support technical issue resolution, quality control, commissioning activities, and project close-out.

2022 – 2025

Maintenance Engineer — Care Solutions — Riyadh, Saudi Arabia

- Executed installation, testing, and commissioning of FCU and VRF systems, including piping, ductwork, copper piping, and equipment connections.
- Installed solar hot water systems including copper piping, primary and secondary circuits, and calorifiers.
- Executed installation of sewage, water supply, firefighting, sprinkler, detection, pump, and associated MEP systems.
- Coordinated site installation and commissioning activities with contractors, technicians, and project teams.
- Ensured installation activities complied with approved drawings, technical specifications, safety requirements, and quality standards.

2019 – 2022

Technical Project Engineer — Integrated International Power Company — Riyadh, Saudi Arabia

- Delivered 1+ MWp of solar PV on-grid projects covering design, procurement, installation, testing, and commissioning.
- Designed solar PV systems using PVsyst and HelioScope, including system configuration and energy-yield analysis.
- Performed solar PV testing and commissioning in accordance with IEC 62446 requirements.
- Prepared technical submittals, feasibility studies, ROI calculations, savings analysis, and project documentation.
- Supported procurement, supplier coordination, site execution, and technical issue resolution throughout project delivery.
- Completed installation, testing, and commissioning of an 850 kWp on-grid solar PV project at SABIC Headquarters.
- Executed solar off-grid systems for traffic monitoring and e-bike mobility applications.

PROJECTS EXECUTED

1. Amazon Jeddah — 2.06 MWp

- Turnkey rooftop solar PV project supporting renewable energy generation and reduction of conventional electricity consumption.
- Involved in technical coordination, project execution, installation, testing, commissioning, and project delivery.

Project Type: Rooftop Solar PV | Turnkey EPC | [Project Details](#)

2. Amazon Riyadh — 2.60 MWp

- Solar PV installation project designed to provide a reliable and efficient renewable energy solution for Amazon's Riyadh facility.
- Supported project execution, technical coordination, installation, testing, commissioning, and delivery activities.

Project Type: Commercial Solar PV | [Project Details](#)

3. Golden Chicken — 5 MWp

- Large-scale solar PV project developed to increase renewable energy generation and reduce dependence on conventional power.
- Supported technical and project execution activities for deployment of the solar PV system.

Project Type: Large-Scale Solar PV | [Project Details](#)

4. Al Jomaih Shell Lubricant Factory — 500 kWp

- Industrial rooftop solar PV installation designed to maximize on-site renewable energy generation and improve energy efficiency.
- Supported technical execution and integration of the solar PV system within an industrial facility environment.

Project Type: Industrial Solar PV | [Project Details](#)

5. Qiddiya Six Flags — 2 MWp

- Solar PV project involving deployment of a robust mounting solution to support reliable and efficient photovoltaic energy generation.
- Contributed to technical coordination and project execution activities for solar infrastructure deployment.

Project Type: Solar PV Infrastructure | [Project Details](#)

6. SABIC Headquarters — 850 kWp

- Delivered an on-grid solar PV project involving system installation, equipment integration, testing, and commissioning.
- Supported technical execution and commissioning activities through project completion and handover.

Project Type: Commercial Solar PV | [Project Details](#)

7. Solar Off-Grid Weather Monitoring Containers

- Engineered stand-alone solar power solutions for remote weather monitoring stations requiring reliable and independent energy supply.
- Supported power provision for remote sensing, data acquisition, communication, and monitoring equipment.

Project Type: Off-Grid Solar System | [Project Details](#)

8. Solar Traffic Monitoring Systems

- Developed solar-powered solutions for traffic monitoring applications in remote and infrastructure locations where conventional grid power was unavailable or impractical.
- Supported implementation of independent renewable power systems for continuous monitoring operations.

Project Type: Off-Grid Solar | Remote Power | [Project Details](#)

9. Solar Off-Grid EV Charging Container

- Designed and delivered an off-grid solar container solution providing an independent renewable power source for electric vehicle charging infrastructure.
- Integrated solar energy generation with an off-grid application to support reliable EV charging in locations with limited grid access.

Project Type: Off-Grid Solar | EV Infrastructure | [Project Details](#)

10. Solar off Grid containers for Weather Monitoring Systems

- Implemented a stand-alone solar energy solution for weather monitoring systems
- Providing dependable power to remote sensing, data acquisition, communication, and monitoring equipment.

Project Type: Off-Grid Solar | EV Infrastructure | [Project Details](#)

11. KAFD Solar Water Heater System

- Solar thermal project at King Abdullah Financial District (KAFD) utilizing solar energy for efficient hot water generation.
- Supported installation and integration of solar thermal components and associated water-heating infrastructure.

Project Type: Solar Thermal | Water Heating | [Project Details](#)

12. STC Parking Area — 100 kWp

- Rooftop solar PV installation utilizing available facility space to generate clean electricity and increase renewable energy contribution.
- Supported solar PV installation and technical execution for efficient utilization of available rooftop infrastructure.

Project Type: Rooftop Solar PV | [Project Details](#)

EDUCATION

B.E, Mechanical Engineering

Osmania University, Hyderabad, India

SKILLS

- Engineering Support
- Project Scheduling
- Reading & Interpreting Engineering Drawings
- Technical Troubleshooting
- Root Cause Analysis
- Quality Analysis
- Solar PV System Design
- Solar Energy Yield Analysis
- PVsyst
- HelioScope
- AutoCAD
- Excel Proficiency
- Solar PV Testing & Commissioning
- Technical Documentation & Reporting

LANGUAGES

English | Arabic